

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(b)	(i)		$K_c = \frac{[\text{CH}_3\text{OH}]}{[\text{CO}][\text{H}_2]^2} \quad (1)$ unit $\Rightarrow \text{dm}^6 \text{ mol}^{-2} \quad (1)$		2		2	1	
		(ii)		$[\text{CH}_3\text{OH}] = K_c \times [\text{CO}] \times [\text{H}_2]^2 \quad (1)$ $[\text{CH}_3\text{OH}] = 0.183 \text{ mol dm}^{-3} \quad (1)$		2		2	2	
	(c)	(i)		$\frac{111.6}{243.6} \times 100 \quad (1)$ 45.8% (1) answer given to 3 sig figs (1)		3		3	1	
		(ii)		moles of $\text{Fe}_2\text{O}_3 = \frac{20\,000\,000}{159.6} = 125313.2 \quad (1)$ moles of $\text{CO}_2 = 3 \times 125313.2 = 375939.6 \quad (1)$ volume of $\text{CO}_2 = \frac{375939.6 \times 8.31 \times 1373}{1.01 \times 10^5} \quad (1)$ volume of $\text{CO}_2 = 42460 \text{ m}^3 \quad (1)$ ecf possible throughout		4		4	3	
				Question 12 total	2	13	2	17	7	0